

Wenxuan Li

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EDUCATION

Cornell University

Incoming PhD Candidate in Electrical and Computer Engineering

Sept 2026 - TBD

Ithaca, US

University of Cambridge

Master of Philosophy in Advanced Computer Science

Oct 2023 - Oct 2025

Cambridge, UK

- Graduated with Distinction; GPA: 4.0/4.0

University of Edinburgh

Bachelor of Science with Honours in Artificial Intelligence and Computer Science

Sep 2019 - May 2023

Edinburgh, UK

- Graduated with First Class Honours; GPA: 3.94/4.0 (Top 2%)

PUBLICATIONS

- **Wenxuan Li***, Chengruidong Zhang*, Huiqiang Jiang*, Yucheng Li, Yuqing Yang, Lili Qiu. ***MTraining***: Distributed Dynamic Sparse Attention for Efficient Ultra-Long Context Training. **MLSys 2026**.
- **Wenxuan Li***, Taiyi Wang*, and Eiko Yoneki. Navigating in High-Dimensional Search Space: A Hierarchical Bayesian Optimization Approach. Preprint arXiv:2410.23148.

INTERN EXPERIENCE

Research Intern (Awarded *Star of Tomorrow*), Microsoft Research Asia

Oct 2025 - Apr 2026

AI Infrastructure Intern, Alibaba Cloud

Aug 2025 - Oct 2025

Research Intern, Microsoft Research Asia (Shanghai)

Dec 2024 - Jun 2025

RESEARCH EXPERIENCE

***MTraining: Distributed Long-Context LLM Training w. Sparse Attention**

Dec 2024 - May 2025

Institute: Microsoft Research Asia (Shanghai)

Research Intern Project

- Empirically and theoretically validated the universal existence of vertical-and-slash (VS) sparse pattern in LLM attention
- Designed hierarchical variant of Ring Attention aligning the computation schedule with the sparse pattern to achieve better workload-balance and computation-communication overlap
- Implemented ideas by custom CUDA and Triton kernels and achieved up to 6× throughput in training Qwen-2.5-3B on 512K-token-long data using 4x 8-GPU nodes with the proposed training strategy

Efficient Distributed LLM Training Simulator

Nov 2025 - Apr 2026

Institute: Microsoft Research Asia

Research Intern Project

- Built flexible and resource-efficient workload generator for distributed training based on automated symbolic execution
- Developed simulation engine for explicit modelling the dynamic dataflow in Expert Parallelism.

DBMS Configuration Tuning via High-dimensional Bayesian Optimization

Spring 2024

Supervisor: Dr Eiko Yoneki (University of Cambridge)

MPhil Dissertation

- Proposed a novel hierarchical BO algorithm (HiBO) for high-dimensional optimization tasks based on adaptive search space partitioning
- Tuned 110 configuration knobs of PostgreSQL with HiBO on SysBench, achieving ~2.25x throughput improvements with limited sample budget
- Validated the algorithm via visualizing dynamics of trust regions and knob-level interpretation of trust regions

Attention-based Knowledge Distillation on Self-supervised Speech Models

Sept 2022 – Apr 2023

Supervisor: Dr Hao Tang (University of Edinburgh)

Undergraduate Dissertation

- Designed layer-structured knowledge distillation scheme based on KL-divergence between attention maps, reducing the teacher model's parameter count by 75%, while maintaining 90% of its accuracy on LibriSpeech test set

OPEN-SOURCE EXPERIENCE

CAMEL-AI.org: Building AI Society with Agents

Jun 2023 - Oct 2023

Mentor: Dr Guohao Li (Eigent AI / CAMEL-AI.org)

Github: <https://github.com/camel-ai/camel>

- Contributed 9 fundamental commits via collaboration to CAMEL, the first multi-agent framework based on LLMs
- Implemented function-calling feature into CAMEL, enabling the agents to call user-customized functions during collaboration (e.g. web search)
- Developed the interface with the LLM inference framework *vLLM*, enabling users to drive agents with open-source LLMs

TECHNICAL PROJECTS

Incremental Optimization of GEMM CUDA Kernels with Nsight Compute (C++ & Python) Aug 2025

FlashAttention from Scratch (C++ & Python) Aug 2024

Pipeline Parallelism from Scratch with PyTorch (Python) Dec 2023 - Jan 2024

InfOS: School's Research Operating System (C++) Jan 2023 - Apr 2023

Rapid-exploring-Random-Tree-based Food Delivery Path Planner (Java) Oct 2021 - Dec 2021

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, Haskell

Packages/Frameworks: CUDA, Nsight System / Compute, Triton, PyTorch, FSDP, TensorFlow, Scikit-Learn, PyTest

Developer Tools: Docker, Linux, Git, Slurm, AWS Toolkits, Azure

TEACHING EXPERIENCE

Tutor for course (*Introduction to*) Machine Learning

Sep 2022 – Dec 2022

Institution: University of Edinburgh

- Participated in the process of preparing the course materials and reflected students' feedback about the course contents to the lecturer
- Hosted 5 hours of tutorials in total, guiding students through questions on machine-learning topics including matrix calculus, computation graphs, and optimization problems.

EXTRACURRICULAR

Student Representative for Year-2 Students

Sep 2020 – May 2021

Institution: University of Edinburgh

- Hosted weekly sessions to collect students' feedbacks about course content and coursework
- Communicated with senior administrative staffs to solve problems reflected by students' feedbacks
- Improved average rate of students satisfaction about study life by ~15% compared to the start of the academic year (indicated by questionnaire)